

SANTA BARBARA, CA, USA

WMO: 723925

Lat: 34.426N

Lon: 119.843W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23190

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.9	3.0	-5.8	2.3	13.2	-3.7	2.8	12.2	9.3	14.7	8.4	14.6	0.3	60	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	28.6	17.5	26.7	17.5	25.1	17.2	20.5	25.1	19.6	24.1	19.0	23.2	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.5	22.6	17.8	12.8	21.5	17.3	12.4	20.9	58.6	25.0	55.9	24.1	53.7	23.2	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.4	7.2	5.8	DB	-0.6	33.8	1.3	2.4	-1.6	35.5	-2.4	36.9	-3.1	38.3	-4.1	40.0
			WB	-1.9	21.4	1.3	1.1	-2.9	22.2	-3.7	22.8	-4.4	23.4	-5.4	24.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	11.8	12.2	13.5	14.5	15.8	17.1	18.8	19.0	18.7	16.9	14.0	11.6
	DBStd	3.46	2.38	2.37	2.31	2.45	2.21	1.70	1.74	1.74	2.24	2.45	2.44	2.30
	HDD10.0	28	9	6	2	1	0	0	0	0	0	0	2	10
	HDD18.3	1225	202	171	152	121	87	46	15	11	21	59	131	211
	CDD10.0	1975	66	68	108	134	179	212	271	281	262	213	123	58
	CDD18.3	128	0	0	1	4	7	7	28	33	32	14	2	0
	CDH23.3	838	12	16	26	49	62	38	80	129	188	180	54	3
	CDH26.7	178	1	1	3	12	18	7	11	17	41	54	12	0
(14) Wind	WSAvg	2.3	1.9	2.3	2.5	2.9	2.8	2.6	2.5	2.4	2.2	2.0	1.8	1.8
(15) Precipitation	PrecAvg	406	82	95	80	24	5	1	1	2	3	13	50	57
	PrecMax	892	311	346	291	146	46	12	10	48	32	61	176	147
	PrecMin	110	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	171	78	85	72	36	11	3	2	8	7	15	51	48
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	25.9	26.5	28.1	29.2	27.5	27.9	28.8	30.3	31.8	28.2	23.6
		MCWB	12.6	13.9	15.1	15.5	15.9	17.5	19.5	20.2	18.5	16.5	14.8	13.1
	2%	DB	22.1	22.2	22.8	24.1	24.5	23.9	25.2	26.3	27.4	27.7	24.5	21.1
		MCWB	12.4	13.3	13.9	14.1	15.6	17.2	19.0	19.5	19.3	16.1	14.1	12.6
	5%	DB	19.7	19.3	20.6	21.8	22.1	22.4	23.8	24.6	25.3	24.9	22.1	19.0
		MCWB	12.3	12.5	13.3	13.9	15.2	16.7	18.5	18.9	18.6	16.0	14.1	12.4
	10%	DB	17.6	17.4	18.5	19.7	20.2	21.1	22.6	23.2	23.7	22.7	20.0	17.4
		MCWB	12.7	12.2	12.9	13.5	14.8	16.4	18.0	18.4	18.2	16.1	13.9	12.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	16.1	16.8	17.4	18.5	19.1	20.9	21.1	21.7	20.7	17.5	16.4
		MCDB	18.8	20.3	22.5	23.9	24.2	24.3	25.2	26.7	26.3	25.3	22.5	19.2
	2%	WB	15.1	15.0	15.5	15.9	17.1	18.1	19.7	20.1	20.5	18.9	16.5	15.2
		MCDB	17.6	18.2	19.6	21.0	21.9	22.4	23.8	25.0	25.0	23.4	20.3	17.8
	5%	WB	14.3	14.2	14.7	15.1	16.3	17.5	19.0	19.3	19.6	17.8	15.7	14.2
		MCDB	16.8	17.1	18.2	19.5	20.4	21.3	23.0	23.8	23.8	21.8	19.4	16.7
	10%	WB	13.5	13.4	14.0	14.4	15.5	16.8	18.3	18.7	18.8	17.0	15.0	13.2
		MCDB	16.2	16.3	17.2	18.3	19.1	20.3	22.1	22.7	22.7	20.7	18.6	16.1
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	10.8	10.0	10.2	9.1	8.2	8.2	8.7	9.7	11.5	12.5	12.2
		MCDBR	16.3	15.3	14.3	14.3	12.8	11.1	10.0	11.1	13.0	16.4	16.3	15.6
	5% WB	MCWBR	9.1	8.3	7.5	7.1	6.3	5.7	4.9	5.4	6.3	8.1	8.8	9.3
		MCWBR	11.0	10.8	10.8	11.6	10.3	9.5	9.1	10.1	10.9	12.0	12.3	11.6
(40) Clear-Sky Solar Irradiance	taub		0.304	0.318	0.326	0.364	0.379	0.365	0.395	0.374	0.363	0.351	0.309	0.311
		taud	2.592	2.566	2.543	2.421	2.396	2.438	2.409	2.435	2.474	2.500	2.599	2.585
	Ebn at Noon		914	935	951	922	906	914	884	901	898	883	901	883
		Edh at Noon	77	88	97	115	120	114	117	112	103	93	76	74
(44) All-Sky Solar Radiation	RadAvg		3.07	3.94	5.31	6.63	7.31	7.60	7.35	6.88	5.89	4.52	3.47	2.78
	RadStd		0.23	0.41	0.34	0.38	0.48	0.35	0.29	0.26	0.24	0.33	0.19	0.22

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		+0.61	N/A	-1.62	+0.68	N/A	N/A	-127	+220	+83

Nomenclature: See separate page